CS 300 – Chapter 11 Lectures

B-Trees

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Key idea Chapter map

Where this sits in CS 300. Chapter 11 is the *shallow-and-wide tree chapter*. **ch. 9** covered AVL and red-black trees as the canonical balanced *binary* trees: height $O(\log_2 n)$, every level branches by 2. Chapter 11 generalises to balanced K*-ary* trees: height $O(\log_K n)$, every level branches by K. The motivating shift is the **memory hierarchy**. When each level of the tree is a disk seek (10 ms), DRAM cache miss (100 ns), or NVMe read (50 μs), reducing height by a factor of $\log_2 K$ is enormous. B-trees are why every modern database, filesystem, and key-value store uses them; the production references at the end of this chapter (§11.7) are dense.

What you add to your toolkit:

- **The minimum-degree parameter.** CLRS calls it t; these notes use *order* K ($K = 2t$ in CLRS terms). The resulting **node-occupancy bounds** ($\lceil K/2 \rceil - 1$ to $K - 1$ keys per non-root node; root exception) are the workhorse invariant the algorithms rely on.
- **Height bound** $h \leq \log_t \frac{n+1}{2}$ — derived from worst-case occupancy. This is why a B-tree of order 100 is only ~ 5 levels deep at $n = 10^9$.
- **Preemptive top-down split.** The single-pass insertion strategy that avoids the upward recursion a naive bottom-up split would force.
- **Preemptive merge.** The mirror: rotate-or-fuse on the way down so the leaf is always safe to delete from.
- **B+ trees.** The production refinement where data lives only at leaves and the leaves form a linked list — this is what your database actually uses, and it's why range queries are cheap.
- **Where B-trees show up in production:** SQLite indexes, PostgreSQL (`btree` access method), MySQL InnoDB (clustered B+ tree on the primary key), Linux ext4 (htree directory entries), Btrfs (copy-on-write B+ trees throughout), git packfile `.idx` (sorted-array, not B-tree — a useful contrast), LSM-trees on top of B-trees (LevelDB, RocksDB, Cassandra).

Mastery checklist — you should be able to, without references:

1. State the B-tree node-occupancy invariant ($\lceil K/2 \rceil - 1 \leq$ keys $\leq K - 1$, root excepted) and sketch why this gives $h = O(\log_K n)$.
2. Run a B-tree search by hand on a 3-level tree of order $K = 4$: identify the in-node scan + descent at each level (§11.2).
3. Walk through a preemptive top-down split insertion: insert into a tree where the descent path contains a full node, and identify how the split propagates the median key to the parent (§11.3).
4. Identify the four CLRS B-TREE-DELETE cases on a worked example: (1) key in leaf, (2) key

in internal node with three subcases 2a/2b/2c, (3) key not in current node with subcases 3a/3b (§11.4–§11.5).

5. Distinguish B-tree from B+ tree: data-at-internal-nodes vs data-at-leaves-only with a linked list of leaves; argue when the B+ form is preferable (range queries, sequential scans, §11.5).
6. Pick the right B-tree order K for a given memory hierarchy. Page size + key+pointer size determine the target fanout (a 4 KB page with 16-byte (key, pointer) slots gives $K \approx 256$; a 1 KB page with 16-byte slots gives $K \approx 64$; §11.1).
7. State why B-trees beat hash tables for filesystem and database indexes (range queries, sequential scans, ordered iteration are all cheap on B-trees and impossible on hash tables; §11.7).

Looking ahead. Chapter 12 covers the *set ADT* in full; B-trees and B+ trees are the disk-side implementation choice when persistence + ordered iteration matter (Java `TreeMap` / C++ `std::set` use red-black; SQLite / PostgreSQL / InnoDB use B+ trees for the same operations). The production-database / systems-programming follow-on direction beyond cs-300 lives in §11.6 (cache-oblivious B-trees, LSM-trees, fractal trees) and §11.7 (the canonical references).

Key idea

Why B-trees exist. AVL and red-black trees (ch. 9) optimize for in-memory comparisons: each node holds one key and two children. But when your tree doesn't fit in RAM – when it lives on disk or across a network – a binary tree is a disaster. Each node traversal is a separate I/O operation. A tree of a billion keys is $\log_2 10^9 \approx 30$ levels deep; that's 30 disk seeks per lookup at roughly 10ms each = 300ms per query. A B-tree of the same size is $\log_{100} 10^9 \approx 4.5$ levels deep. Fatter nodes, fewer levels, fewer I/Os. *B-trees are what happens when you let cache lines or disk blocks dictate your node size.*

1 11.1 B-tree Structure

The definition

Definition —

B-tree of order K. A search tree where every node holds at most $K-1$ keys and at most K children. Subject to:

- All keys in a node are stored in sorted order.
- An internal node with n keys has exactly $n+1$ children.
- Between two adjacent keys, the child subtree holds keys strictly between them; outside the first or last key, the outer subtrees hold keys strictly less / greater.
- All leaves sit at the same depth. (This is the balance condition.)
- **Minimum-occupancy floor.** Every internal node *except the root* holds at least $\lceil K/2 \rceil - 1$ keys (equivalently, at least $\lceil K/2 \rceil$ children). The root is exempt: it may carry as few as 1 key (or 0 keys when the tree is empty) and as many as $K-1$.
- All keys are distinct.

Aside

Different textbooks use subtly different definitions of "order" – some count maximum children (K), some count maximum keys ($K-1$), some use a *minimum* degree t with $t-1$ to $2t-1$ keys per node. These notes use "order K = max children." In the CLRS convention, $K = 2t$, so the minimum-keys floor $\lceil K/2 \rceil - 1$ becomes $t-1$ and the maximum is $2t-1$. Translate carefully if you cross-reference CLRS or Knuth.

Aside

Root exception. The minimum-keys floor is the workhorse invariant for most of the algorithms in this chapter — preemptive split, preemptive merge, fusion, rotation all assume that any node we touch except the root has $\geq \lceil K/2 \rceil - 1$ keys. The root is special on both ends: when the tree is small the root may carry just one key (or be a leaf), and when the tree is empty the root pointer can be `nullptr`. *Root-split* (§11.3) and *root-fusion* (§11.4) are the two events that propagate height changes through this exception.

The 2-3-4 tree – a B-tree of order 4

Order-4 B-trees are the standard teaching example. A node holds:

- 1, 2, or 3 keys – called **2-node**, **3-node**, or **4-node**.
- Correspondingly 2, 3, or 4 children (for internal nodes).

A common convention labels node keys as A, B, C and children as `left`, `middle1`, `middle2`, `right`:

Node type	Keys used	Children used
2-node	A	`left`, `middle1`
3-node	A, B	`left`, `middle1`, `middle2`
4-node (full)	A, B, C	`left`, `middle1`, `middle2`, `right`

Key idea

2-3-4 trees are a finger exercise. Real B-trees in databases use orders in the hundreds or thousands – typically whatever fits in one disk page (4KB–16KB) or one L2 cache line (a few hundred integers). The *algorithms* are identical to the order-4 version; only the array sizes change.

C++ node skeleton

```
template<int Order>   // e.g. Order=4 for 2-3-4 trees
struct BTreeNode {
    int numKeys = 0;
    int keys[Order - 1];
    BTreeNode* children[Order] = {};     // null for leaves
    BTreeNode* parent = nullptr;

    bool isLeaf() const { return children[0] == nullptr; }
    bool isFull() const { return numKeys == Order - 1; }
};

template<int Order>
```

```
struct BTree {
    BTreeNode<Order>* root = nullptr;
};
```

Helper primitives

The insert / remove / rotate / fuse listings later in this chapter call into a small set of in-node primitives. We collect their signatures + brief implementations here so the chapter listings are runnable-shape rather than illustrative-only:

```
// Linear-scan key lookup inside one node. Returns -1 on miss.
template<int Order>
int indexOfKey(BTreeNode<Order>* n, int key) {
    for (int i = 0; i < n->numKeys; ++i)
        if (n->keys[i] == key) return i;
    return -1;
}
template<int Order>
bool containsKey(BTreeNode<Order>* n, int key) {
    return indexOfKey(n, key) != -1;
}

// Pick the child whose subtree must contain `key` (caller guarantees
// the key is not in `n` itself).
template<int Order>
BTreeNode<Order>* btreeChildForKey(BTreeNode<Order>* n, int key) {
    int i = 0;
    while (i < n->numKeys && key > n->keys[i]) ++i;
    return n->children[i];
}

// Insert `key` into a non-full leaf. Shifts existing keys right.
template<int Order>
void btreeInsertIntoLeaf(BTreeNode<Order>* n, int key) {
    int i = n->numKeys - 1;
    while (i >= 0 && n->keys[i] > key) {
        n->keys[i+1] = n->keys[i]; --i;
    }
    n->keys[i+1] = key; ++n->numKeys;
}

// Insert (key, leftChild, rightChild) into a non-full internal node,
// preserving sort order. Used by btreeSplit on the parent.
template<int Order>
void btreeInsertKeyWithChildren(BTreeNode<Order>* p, int key,
                                BTreeNode<Order>* L, BTreeNode<Order>* R) {
    int i = p->numKeys - 1;
    while (i >= 0 && p->keys[i] > key) {
        p->keys[i+1]      = p->keys[i];
        p->children[i+2]  = p->children[i+1];
        --i;
    }
    p->keys[i+1]     = key;
```

```
    p->children[i+1] = L;
    p->children[i+2] = R;
    ++p->numKeys;
    L->parent = R->parent = p;
}

// Drop key at index `idx`, shifting neighbours left.
template<int Order>
void removeKey(BTreeNode<Order>* n, int idx) {
    for (int i = idx; i < n->numKeys - 1; ++i)
        n->keys[i] = n->keys[i+1];
    --n->numKeys;
}

// Smallest key in the subtree rooted at `n` (leftmost descendant).
template<int Order>
int btreeGetMinKey(BTreeNode<Order>* n) {
    while (!n->isLeaf()) n = n->children[0];
    return n->keys[0];
}
```

The split / rotate / merge / fuse / remove listings later treat the helpers above as black-box primitives. Production B-tree code factors them similarly — the algorithm-level listings stay readable, the in-node bookkeeping lives in one place.

The CLRS minimum-degree convention — t vs K

Aside

$t \leftrightarrow K$ **bridge.** CLRS Ch. 18 parameterises B-trees by a *minimum degree* $t \geq 2$: every internal node except the root has at least $t-1$ keys and at most $2t-1$ keys (equivalently, between t and $2t$ children). Order K as defined above relates by $K = 2t$, so:

- Min keys per non-root: $\lceil K/2 \rceil - 1 = t - 1$.
- Max keys per node: $K - 1 = 2t - 1$.
- Min children per non-root: $\lceil K/2 \rceil = t$.
- Max children per node: $K = 2t$.

The 2-3-4 tree has $t = 2$, $K = 4$: 1 to 3 keys, 2 to 4 children. Read *order* as "what's the maximum branching?" and *minimum degree* as "what's the slack-to-merge threshold?"; both encode the same bounds.

Height analysis

A B-tree of order K with n keys has height $h = O(\log_K n)$. For $K = 100$ and $n = 10^9$, that's $\log_{100} 10^9 \approx 4.5$ – roughly 5 node accesses per lookup. Contrast with a binary tree's $\log_2 10^9 \approx 30$.

That factor-of-six reduction is the whole point. On disk it translates directly to query latency.

Definition —

Height bound (CLRS Theorem 18.1). If $n \geq 1$ keys live in a B-tree T of minimum degree $t \geq 2$, then

$$h \;\leq\; \log_t \frac{n+1}{2}.$$

Sketch. Worst case: every node hits the minimum-occupancy floor (root has 1 key, every other node has $t-1$ keys). Counting nodes per level:

- Level 0 (root): 1 node.
- Level 1: ≥ 2 nodes (root has 2 children minimum).
- Level 2: $\geq 2t$ nodes (each level-1 node has $\geq t$ children).
- Level $i \geq 1$: $\geq 2t^{i-1}$ nodes.

Each node holds at least $t-1$ keys, plus the root contributes 1 key, so

$$n \;\geq\; 1 + (t-1)\sum_{i=1}^{h} 2t^{i-1} \;=\; 1 + 2(t-1)\cdot\frac{t^h-1}{t-1} \;=\; 2t^h - 1.$$

Solving: $t^h \leq (n+1)/2$, hence $h \leq \log_t \frac{n+1}{2}$. **Q.E.D.**

The disk-access cost model

Definition —

Disk-access model (DAM). The B-tree cost model of record. Memory is partitioned into a small *internal* buffer (DRAM, M words) and an unbounded *external* store (disk, page size B words). Computation on internal data is free; the only cost counted is **block transfers** between the two. A B-tree of order K tuned so one node = one disk page achieves $\Theta(\log_K n) = \Theta(\log_B n)$ block transfers per search / insert / delete — and that's the whole point of the data structure.

Aside

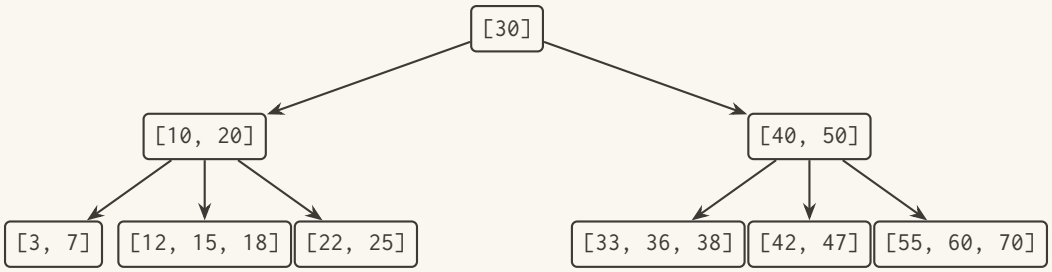
Concrete sizing. Pick K so one node fills the underlying block size. A few worked examples:

- **1 KB page, 8-byte keys + 8-byte child pointers** $\Rightarrow$ each (key, pointer) slot is 16 bytes, $K \approx 1024/16 = 64$.
- **4 KB page, same slot size** $\Rightarrow K \approx 256$. SQLite's default.
- **16 KB page** (MySQL InnoDB default, PostgreSQL configurable) $\Rightarrow K \approx 1024$, height ≤ 4 for $n = 10^{12}$.
- **64-byte cache line, 16-byte slots** $\Rightarrow K \approx 4$. The 2-3-4 register comes back when your "page" is a cache line — relevant for in-memory B-trees in modern OLTP engines.

The take-away: K is set by hardware geometry, not by any algorithmic constraint. Pick the largest K that keeps a node in one block.

A 3-level B-tree of order 4 — the running picture

The structural invariants are easier to internalise on a worked diagram. Below is a 3-level B-tree of order $K = 4$ (so 1–3 keys per node, 2–4 children, leaves at the same depth):



Read off each invariant on the picture:

- **Sorted within node.** Every node's keys are left-to-right ascending: `[10, 20]`, `[12, 15, 18]`, `[33, 36, 38]`.
- $n + 1$ **children.** The root has 1 key, 2 children. `[10, 20]` has 2 keys, 3 children. The 4-node `[33, 36, 38]` would have 4 children if it were internal — here it's a leaf.
- **Subtree key-ranges.** Under `[10, 20]`: left child is < 10 (`[3, 7]`), middle child is between 10 and 20 (`[12, 15, 18]`), right child is > 20 (`[22, 25]`).
- **All leaves at the same depth.** Every leaf sits at depth 2.
- **Min-occupancy floor.** Order $K = 4 \Rightarrow \lceil 4/2 \rceil - 1 = 1$ key minimum per non-root. Every non-root node holds ≥ 1 key. Root exception: `[30]` has exactly 1 key (legal because it's the root).

This tree carries 16 keys at depth 3. With $K = 100$ on the same 3 levels, a B-tree carries up to $100^3 = 10^6$ keys — the same shape, vastly more capacity.

Example

Where B-trees actually live.

- **Database indexes**: MySQL (InnoDB), PostgreSQL, SQL Server, Oracle all use B+ trees (a B-tree variant) for their default index structures.
- **Filesystems**: ext4, NTFS, HFS+, APFS, Btrfs (the "B" is for B-tree), XFS – all B-tree-based for directory and extent management.
- **Key-value stores**: BoltDB, LMDB use B+ trees.
- **Version control**: Git's packfile index uses a B-tree-like structure for fast object lookup.

You won't "implement a B-tree for production" very often, but you'll read about them in every system's architecture doc.

2 11.2 Search

The algorithm

B-tree search is binary search *within* a node combined with tree descent. At each node:

1. Linear-scan (or binary-search, for larger orders) the node's keys for the target.
2. If found, return the node.
3. Otherwise, find the child whose key range contains the target, and recurse.

For the 2-3-4 case, the child selection reduces to a short comparison chain:

Condition	Descend to
key $<$ node.A	`left`
node has 1 key, or key $<$ node.B	`middle1`
node has 2 keys, or key $<$ node.C	`middle2`
otherwise	`right`

```
BTreeNode* btreeSearch(BTreeNode* node, int key) {
    if (!node) return nullptr;

    // Check keys in this node.
    if (node->numKeys >= 1 && node->keys[0] == key) return node;
    if (node->numKeys >= 2 && node->keys[1] == key) return node;
    if (node->numKeys >= 3 && node->keys[2] == key) return node;

    // Descend to the appropriate child.
    if (key < node->keys[0])
        return btreeSearch(node->children[0], key);     // left
    if (node->numKeys == 1 || key < node->keys[1])
        return btreeSearch(node->children[1], key);     // middle1
    if (node->numKeys == 2 || key < node->keys[2])
        return btreeSearch(node->children[2], key);     // middle2
    return btreeSearch(node->children[3], key);         // right
}
```

B-TREE-SEARCH — the general form

The 2-3-4-special listing above unrolls the in-node comparisons by hand. CLRS Ch. 18.1 gives the general recursive form that works for any order K: scan the keys for the smallest one $\geq$ the target, either return on a hit or descend into the matching child.

```
template<int Order>
BTreeNode<Order>* btreeSearchGeneric(BTreeNode<Order>* node, int key) {
    if (!node) return nullptr;
    int i = 0;
    while (i < node->numKeys && key > node->keys[i]) ++i;
    if (i < node->numKeys && node->keys[i] == key) return node;
    if (node->isLeaf()) return nullptr;
    return btreeSearchGeneric(node->children[i], key);
}
```

Aside

Linear vs binary in-node scan. The listing above uses a linear scan (`while (key > keys[i])`). For small K that's optimal — branch prediction and the linear-scan-fits-in-a-cache-line property dominate the asymptotics. CLRS uses linear in pseudocode for clarity. Production code with K in the hundreds or thousands swaps in a binary search inside the node:

```
int lo = 0, hi = node->numKeys;
```

```
while (lo < hi) {
    int mid = (lo + hi) / 2;
    if (node->keys[mid] < key) lo = mid + 1;
    else                       hi = mid;
}
```

giving $O(\log K)$ per-node CPU. The choice doesn't change the disk-cost calculus — both still touch one node per level.

Cost — CPU vs disk decomposition

Cost dimension	Linear in-node scan	Binary in-node scan
Disk reads (block transfers)	$O(\log_K n)$	$O(\log_K n)$
CPU comparisons / node	$O(K)$	$O(\log K)$
Total CPU work	$O(K \log_K n)$	$O(\log K \cdot \log_K n) = O(\log n)$
Memory accesses	1 per node-step	1 per node-step

The two takeaways:

- **Disk cost is independent of in-node search style.** Both linear and binary read the same number of blocks — one per level. That's what fanout buys you.
- **CPU cost equals balanced-binary-tree cost** once you switch to binary in-node search: $O(\log K \cdot \log_K n) = O(\log_2 n)$. So a B-tree is *never* CPU-slower than a binary tree (on identical data) and is dramatically faster on disk-bound workloads.

Key idea

Big-O hides the real win. A binary tree and a B-tree both do $O(\log n)$ comparisons, but the B-tree does them with one-tenth the *pointer dereferences*. When each pointer dereference is a cache miss (or a disk seek), that's a 10× speedup, even though the asymptotic complexity is identical.

3 11.3 Insertion

The big picture

New keys always go into leaf nodes. The challenge: leaves might already be full. Strategy:

Key idea

Preemptive split. As we descend from the root to the target leaf, *split any full node we encounter along the way*. This guarantees the target leaf has room. Splits never cascade up, because every ancestor we visited is already non-full (we would have split it if not).

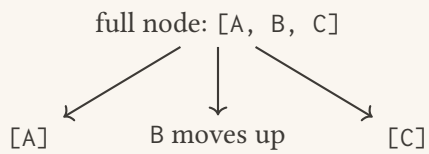
Aside

Why preemptive top-down (and not retroactive bottom-up)? The "obvious" algorithm is: descend to the leaf, insert, and if the leaf overflows, split it and propagate the median key up to the parent — recursively splitting ancestors that overflow. That's Knuth's classical algorithm. It works, but it requires either a **second pass** from leaf back up to root or **parent pointers** on every node so the

recursion can return its way up.
CLRS Ch. 18.2 (and these notes) take the dual view: split any full node *before* descending into it. The invariant "the current node has room" is preserved at every step of the recursion, so a single downward pass suffices. The cost is splitting some nodes that might not have actually needed it (you could've descended through a full node without overflowing anything below) — but the algorithmic simplification is enormous: no upward recursion, no parent pointers required, no second pass. The "wasted work" is bounded: at most one split per level of the tree, so $O(\log_K n)$ extra work per insert in the worst case.

The split operation

A full node (with its maximum $K-1$ keys) is split into two nodes of $(K/2-1)$ keys each, and the middle key is promoted to the parent. For 2-3-4 trees:



Keys A and C become two new sibling nodes; key B moves into the parent, with pointers to the new siblings. The parent must have room (thanks to preemptive split).

```
BTreeNode* btreeSplit(BTree& t, BTreeNode* node) {
    if (!node->isFull()) return nullptr;
    BTreeNode* parent = node->parent;
    BTreeNode* left  = new BTreeNode{1, {node->keys[0]},
                                     {node->children[0], node->children[1]},
                                     parent};
    BTreeNode* right = new BTreeNode{1, {node->keys[2]},
                                     {node->children[2], node->children[3]},
                                     parent};

    if (parent) {
        btreeInsertKeyWithChildren(parent, node->keys[1], left, right);
    } else {
        parent = new BTreeNode{1, {node->keys[1]}, {left, right}, nullptr};
        t.root = parent;
        left->parent = right->parent = parent;
    }
    return parent;
}
```

B-TREE-SPLIT-CHILD — the CLRS form

The listing above is hard-coded to order 4. CLRS Ch. 18.2's B-TREE-SPLIT-CHILD generalises to any order. The invariant: the parent p is non-full (so it has room for one more key + child pointer), and the child to split is at index i. The child is full ($K-1$ keys); after the split, we get two siblings each with $\lfloor (K-1)/2 \rfloor$ keys, and the median key moves up to p.

```
template<int Order>
```

```
void btreeSplitChild(BTreeNode<Order>* p, int i) {
    constexpr int t = Order / 2;          // CLRS minimum-degree t
    BTreeNode<Order>* y = p->children[i];  // the full child
    BTreeNode<Order>* z = new BTreeNode<Order>{};

    // z gets y's right half (keys [t .. 2t-2], children [t .. 2t-1]).
    z->numKeys = t - 1;
    for (int j = 0; j < t - 1; ++j)
        z->keys[j] = y->keys[j + t];
    if (!y->isLeaf()) {
        for (int j = 0; j < t; ++j) {
            z->children[j] = y->children[j + t];
            z->children[j]->parent = z;
            y->children[j + t] = nullptr;
        }
    }
    y->numKeys = t - 1;                   // y keeps left half

    // Make room in p for the new key + child pointer at slot i.
    for (int j = p->numKeys; j > i; --j)
        p->children[j + 1] = p->children[j];
    p->children[i + 1] = z;
    z->parent = p;
    for (int j = p->numKeys - 1; j >= i; --j)
        p->keys[j + 1] = p->keys[j];
    p->keys[i] = y->keys[t - 1];          // promote median
    ++p->numKeys;
}
```

B-TREE-INSERT-NONFULL — the recursive descent

CLRS Ch. 18.2 splits the insert API in two: `btreeInsert` handles the root-split-as-only-height-growth event; everything else goes through `btreeInsertNonFull`, which assumes the node it's called on already has room. This is the cleanest expression of the preemptive-split invariant.

```
template<int Order>
void btreeInsertNonFull(BTreeNode<Order>* node, int key) {
    int i = node->numKeys - 1;
    if (node->isLeaf()) {
        // Shift keys right, place new key in sorted slot.
        while (i >= 0 && node->keys[i] > key) {
            node->keys[i + 1] = node->keys[i]; --i;
        }
        node->keys[i + 1] = key;
        ++node->numKeys;
    } else {
        // Find child whose subtree contains key.
        while (i >= 0 && node->keys[i] > key) --i;
        ++i;
        // Preemptive split: if descent target is full, split first.
        if (node->children[i]->isFull()) {
            btreeSplitChild(node, i);
            // After split, decide which of the two halves to descend.
```

```
            if (key > node->keys[i]) ++i;
        }
        btreeInsertNonFull(node->children[i], key);
    }
}

template<int Order>
void btreeInsertCLRS(BTree<Order>& t, int key) {
    if (!t.root) { t.root = new BTreeNode<Order>{}; }
    if (t.root->isFull()) {
        // Root split: brand-new root, height grows by 1.
        BTreeNode<Order>* s = new BTreeNode<Order>{};
        s->children[0] = t.root;
        t.root->parent = s;
        t.root = s;
        btreeSplitChild(s, 0);
    }
    btreeInsertNonFull(t.root, key);
}
```

The split-and-shift bookkeeping is fiddly because the arrays are fixed-size — everything is in-place index arithmetic. The algorithmic shape is identical to the order-4 `btreeSplit` above; only the loop bounds change.

Special case: splitting the root

When the root itself is full and we start inserting, the split creates a brand-new root with a single key. **This is the only way a B-tree grows in height.** Every other operation happens at the leaf level or via splits that push keys sideways to a non-full ancestor.

Key idea

B-trees grow from the *top* (via root splits), unlike binary search trees which grow from the bottom. That's why all leaves stay at the same depth – depth increases uniformly when the root splits.

Insert into a leaf

Once we've descended (splitting along the way) to the target leaf, the leaf is guaranteed non-full. Insert the new key in sorted position:

Condition (non-full leaf)	Action
key equals existing key	no-op (duplicates forbidden)
key $<$ node.A	shift existing keys right, place new key at A
key $<$ node.B (or no B)	shift $B \to C$ if needed, place new key at B
otherwise	place new key at C

The full insert

```
BTreeNode* btreeInsert(BTreeNode* node, int key) {
    if (containsKey(node, key)) return nullptr;   // duplicates rejected

    if (node->isFull()) node = btreeSplit(node);  // preemptive split
```

```
    if (!node->isLeaf()) {
        BTreeNode* child = btreeChildForKey(node, key);
        return btreeInsert(child, key);
    }
    btreeInsertIntoLeaf(node, key);
    return node;
}
```

Cost

- Descent: $O(\log_K n)$ levels.
- At each level: at most one split, $O(K)$ work.
- Total: $O(K \log_K n)$. For fixed K, that's $O(\log n)$.

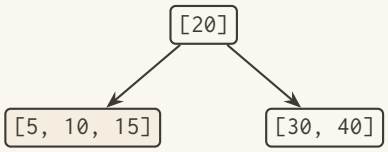
Gotcha

The "lazy" alternative. Instead of preemptive split, you can insert into the leaf first, and *then* split if the leaf became full – cascading splits upward. That's the classical Knuth algorithm and is slightly more space-efficient (nodes can fill up more before being split). Preemptive split is preferred in teaching because it's a single downward pass – no recursion up the tree.

Preemptive split during descent — before / after

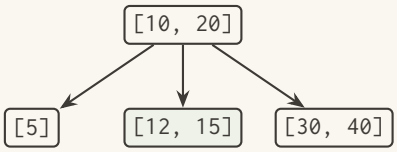
The order-4 split TikZ above shows the structural rearrangement of one split. The diagram below shows the full preemptive-descent *move*: we are inserting a new key, the descent path passes through a full node, and we split that node *before* entering it so the parent (which is guaranteed non-full by induction) absorbs the median.

Before (descend into full child)



target leaf for key=12 is the full `[5,10,15]` *node*

After (preemptive split)



median 10 promoted; descend into non-full `[12,15]` *(or* `[5]`*) and insert*

The full `[5,10,15]` child is split *before* we descend into it: median 10 moves up to the parent (which had room because **we already split it on the way down if it had been full**), and we now have two non-full children to choose from. The new key 12 goes into `[12, 15]` via `btreeInsertNonFull`.

Example

Insert sequence into an empty B-tree of order 4. We insert the keys 10, 20, 30, 40, 5, 15, 25, 35, 45 in order, watching the preemptive-split invariant in action.
Frame 1 (after 10, 20, 30): root is a single 3-key node.

`[10, 20, 30]`

Frame 2 (insert 40 — root is full). CLRS variant: `btreeInsertCLRS` sees `t.root->isFull()`, allocates a new root, makes the old root its only child, then calls `btreeSplitChild(newRoot, 0)`. The median 20 moves up. Then `btreeInsertNonFull(newRoot, 40)` descends right and inserts:

```
                [20]
                /  \
          [10]  [30, 40]
```

Frame 3 (insert 5): root `[20]` not full; descend left into `[10]` (not full); insert in sorted order.

```
                [20]
                /  \
        [5, 10]  [30, 40]
```

Frame 4 (insert 15): root not full; descend left; `[5, 10]` not full; insert. Then descend right path with the **next** insertion (25): `[5, 10, 15]` would be full on the way; we don't touch it because 25 goes right.

```
                [20]
                /  \
    [5, 10, 15]  [25, 30, 40]
```

(Inserting 25 into `[30, 40]` just shifts in place; that node is still non-full at 3 keys.)

Frame 5 (insert 35 — right child is full). Descent reaches the root `[20]` (not full) and prepares to descend right. The right child `[25, 30, 40]` is full, so `btreeSplitChild(root, 1)` runs first: median 30 moves up to the root, splitting the child into `[25]` and `[40]`. **Then** we descend into the appropriate half (35 is between 30 and 40, so we land in `[40]`) and insert:

```
               [20, 30]
               /  |  \
    [5, 10, 15]  [25]  [35, 40]
```

Frame 6 (insert 45): root `[20, 30]` not full; descend right into `[35, 40]`; not full; insert.

```
               [20, 30]
               /  |  \
    [5, 10, 15]  [25]  [35, 40, 45]
```

Two splits in 9 inserts; both happened *before* the descent that needed them. No split ever propagated upward retroactively.

4 11.4 Rotations and Fusion

Removal is the hard half of any balanced tree. For B-trees, the two tools are **rotation** (borrow a key from a sibling) and **fusion** (merge three nodes into one – the inverse of split).

The CLRS 4-case deletion taxonomy

CLRS Ch. 18.3 organises B-tree deletion into four cases, branching on *where* the key sits and *whether* the path to it is "safe to descend" (every non-root node has $\geq t$ keys, i.e., one above the floor, so we can remove a key without underflowing). The taxonomy is the canonical mental model for the deletion algorithm:

Definition —

B-TREE-DELETE cases (CLRS 18.3). Let $t = \lceil K/2 \rceil$ (the minimum-degree parameter). To delete `key` from the B-tree rooted at `r`:

- **Case 1.** `key` is in node x *and* x is a leaf. Just delete `key` from x (shift neighbours left). Safe by the descent invariant: x had $\geq t$ keys before this operation.

- **Case 2.** `key` is in internal node x at slot i. Three subcases on the children $y = x \to$`children[i]` (predecessor side) and $z = x \to$`children[i+1]` (successor side):
  - **2a.** y has $\geq t$ keys. Find the *predecessor* (max key in subtree of y), swap into x's slot i, and recursively delete the predecessor from y.
  - **2b.** y has $t - 1$ keys but z has $\geq t$ keys. Mirror: use the *successor* from z.
  - **2c.** Both y and z have $t - 1$ keys. **Fuse** them with the separator key `key` in between, producing one node with $2t - 1$ keys, then recursively delete `key` from the fused node. (`key` is now in that node; recursion lands in case 1 or recurses further.)
- **Case 3.** `key` is *not* in node x; the descent must continue into child $c_i = x \to$`children[i]`. Before descending, ensure c_i has $\geq t$ keys (so the recursive call inside c_i's subtree can complete safely):
  - **3a.** c_i has $t - 1$ keys but an immediate sibling has $\geq t$ keys. **Rotate** a key from the sibling through x into c_i.
  - **3b.** c_i has $t - 1$ keys and both immediate siblings have $t - 1$ keys. **Fuse** c_i with one sibling and the separator key from x.

  Then descend into the (now-safe) child and recurse.
- **Special case:** the root drops to 0 keys after a fusion. In that case, the tree shrinks: the new root becomes the (unique) child. **This is the only way a B-tree's height shrinks.**

Aside

Why preemptive merge during descent? Same engineering move as preemptive split for insertion. A naive "delete the leaf key, then fix any underflow on the way back up" algorithm requires either parent pointers or a return-from-recursion second pass. By preemptively rotating-or-fusing on the way down (case 3a / 3b *before* entering the child), we maintain the invariant "every node we touch has $\geq t$ keys" — the deletion at the leaf can proceed without underflowing, no upward recursion needed. As with preemptive split, we do a small amount of work (some rotations or fuses) that might not have been strictly necessary, in exchange for a single-pass algorithm with no parent pointers.

Rotation

Definition —

Rotation (B-tree sense). Move one key from a source node, *through the parent*, into a sibling. The source loses a key, the sibling gains a key, and the parent key that was between them is swapped out for a key from the source.

Key idea

This is *not* the same as the AVL/red-black rotation. AVL rotations re-parent subtrees to restore a height balance. B-tree rotations **redistribute keys across sibling nodes** – the tree shape doesn't change, only which key lives in which node.

Right rotation on `node`

Takes a key from `node`'s left sibling, moves the parent separator key into `node`, and pushes the left sibling's rightmost key up to fill the separator slot. Result: the left sibling loses one key, `node` gains one.

Left rotation on node

Mirror image: takes a key from node's right sibling.

```
// Illustrative form. A "rotation on node X" in this context means
// "give X one more key by borrowing from a sibling through the parent."
void btreeRotateLeft(BTreeNode* node) {
    BTreeNode* leftSib = getLeftSibling(node);
    int parentKey = getParentKeyLeftOfChild(node->parent, node);
    addKeyAndChild(leftSib, parentKey, node->children[0]);
    setParentKeyLeftOfChild(node->parent, node, node->keys[0]);
    removeKey(node, 0);
}
```

Fusion

Definition —

Fusion. Merge a parent separator key with two single-key adjacent siblings into one fused 3-key node. *The inverse of split.*

Before fusion	$\rightarrow$	After fusion
`[L]` $\cdot$ (parent sep M) $\cdot$ `[R]`	$\rightarrow$	`[L, M, R]`

The parent loses a key (and a child pointer); the fused node has three keys. Fusion is used when neither sibling has a spare key to rotate.

Root fusion – a special case

When the root has 1 key and both its children have 1 key each, no rotation is possible and ordinary fusion would empty the parent. Instead, the three keys are combined into a *new root* of three keys, and the tree shrinks in height by 1:

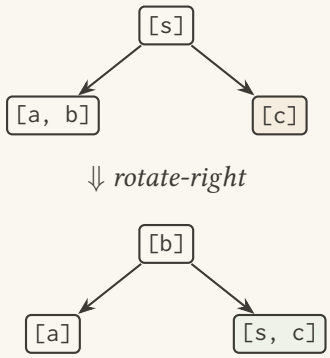
Key idea

Root fusion is the only way a B-tree shrinks in height, mirroring how root split is the only way it grows. Every other operation happens at leaf level or via local restructuring.

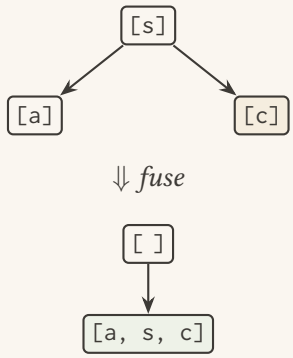
Picture: rotation from sibling vs. fusion with sibling

Cases 3a and 3b of the deletion taxonomy decide between the two deletion-prep strategies. The diagrams below show both moves on the same starting configuration — the deficient child `[c]` has $t - 1 = 1$ key (order $K = 4$), and we need to ensure it has at least $t = 2$ before descending into it.

Case 3a: rotation from sibling
sibling `[a, b]` *has* $\geq t$ *keys*



Case 3b: fusion with sibling
sibling `[a]` *also has* $t - 1$ *keys*



Reading the two cases. The shaded child is the one we're about to descend into; we must raise its key count above the floor first.

- **Case 3a (left).** Sibling `[a, b]` has $\geq t$ keys. **Rotate-right**: b stays in the left sibling, a? No — the parent separator s comes down into c, and the sibling's rightmost key b moves up to take s's place. (Mirror of left-rotation when the spare key is in the right sibling.) The child gains one key, the sibling loses one, the parent's key changes but its count stays the same.
- **Case 3b (right).** Both siblings have only $t - 1$ keys; no rotation possible. **Fuse** the parent's separator s with both children into one node of $2t - 1$ keys. The parent loses one key + one child pointer. If the parent was the root and is now empty, the fused node becomes the new root — the only way a B-tree shrinks in height (mirror of root-split being the only way it grows).

Visualizing split vs. fusion

Operation	Effect on height	Inverse of
Root split	+1	Root fusion
Internal split	0	Internal fusion
Right rotation	0	Left rotation (roughly)

The symmetry is beautiful: *every modification to a B-tree is either a rotation (constant-cost redistribution) or a split/fuse (constant-cost restructuring).* Both are strictly local.

5 11.5 Removal

The same dual-pass idea as insertion

Insertion used *preemptive split*: any full node seen on the way down gets split before we touch it, guaranteeing the target leaf has room.

Removal uses the mirror: **preemptive merge**. Any single-key non-root node seen on the way down gets merged (via rotation or fusion) to have ≥ 2 keys before we touch it. That guarantees the target leaf has ≥ 2 keys and can give one up.

Key idea

The parallel is exact. Inserting needs non-full leaves; removing needs non-minimal leaves. Both algorithms do a single downward pass, maintaining the invariant that the current node is safe to modify.

The merge procedure

Given a 1-key non-root node whose parent has ≥ 2 keys, merge as follows:

```
BTreeNode* btreeMerge(BTreeNode* node) {
    BTreeNode* leftSib  = getLeftSibling(node);
    BTreeNode* rightSib = getRightSibling(node);

    // Preference 1: rotate from a sibling with a spare key.
    if (leftSib && leftSib->numKeys >= 2) {
        btreeRotateRight(leftSib);   // one of leftSib's keys flows into node
    } else if (rightSib && rightSib->numKeys >= 2) {
        btreeRotateLeft(rightSib);
    } else {
        // Preference 2: fuse with a 1-key adjacent sibling.
        if (!leftSib) node = btreeFuse(node, rightSib);
        else          node = btreeFuse(leftSib, node);
    }
    return node;
}
```

Leaf case vs. internal case

Leaf removal. Once preemptive merging has guaranteed the leaf has ≥ 2 keys, just splice the key out and shift neighbors left.

Internal removal. Replace the key with its *successor* (the minimum key in its right subtree) – just like BST removal. But now we have to recursively remove the successor, which *is* in a leaf. The remove call:

1. Find and remember the successor key.
2. Recursively remove the successor (which triggers a leaf removal, safe after merging).
3. Swap the saved successor key into the original internal node's position.

```
bool btreeRemove(BTree& t, int key) {
    // Special case: single-key leaf root.
    if (t.root->isLeaf() && t.root->numKeys == 1 && t.root->keys[0] == key) {
        delete t.root; t.root = nullptr; return true;
    }

    BTreeNode* cur = t.root;
    while (cur) {
        // Preemptive merge: no 1-key non-root nodes on our path.
        if (cur->numKeys == 1 && cur != t.root)
            cur = btreeMerge(cur);
```

```
        int idx = indexOfKey(cur, key);
        if (idx != -1) {
            if (cur->isLeaf()) {
                removeKey(cur, idx);
                return true;
            }
            // Internal: swap with min of right child, then remove the min.
            BTreeNode* rightChild = cur->children[idx + 1];
            int replacement = btreeGetMinKey(rightChild);
            btreeRemove(t, replacement);
            btreeKeySwap(t.root, key, replacement);
            return true;
        }
        cur = btreeNextNode(cur, key);   // descend toward the target
    }
    return false;   // not found
}
```

Cost

Same as insertion: $O(K \log_K n) = O(\log n)$. Single downward pass, constant work per level.

Why preemptive merging is elegant

Key idea

A naive removal needs recursion back up the tree when a leaf underflows. Preemptive merging eliminates upward recursion entirely by strengthening every node *before* it gets weakened. It's the same engineering move as preemptive split on insert – trade a tiny bit of extra work (merging nodes that might not have needed it) for a much simpler one-pass algorithm.

B+ trees – the production refinement

Real databases and filesystems use **B+ trees**, not plain B-trees. The variant changes *where data lives* and adds a linked-list-of-leaves structure on top. The algorithms (search, insert, preemptive split, preemptive merge, rotate, fuse) adapt directly — the only change is that internal nodes hold routing keys but no values, and leaves chain to their right neighbours.

Definition —

B+ tree. A B-tree where:

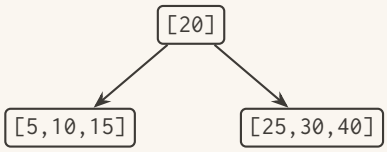
- **Data only at leaves.** Every record (key, value) pair lives in a leaf. Internal nodes hold *routing keys* only — copies of leaf keys used to decide which subtree to descend into. A routing key in an internal node is duplicated in some leaf below it (this is the trade for higher fanout).
- **Linked leaves.** Each leaf has a `next` pointer to its right sibling at the leaf level (some implementations also chain backward). Range scans walk the linked list rather than re-descending from the root.
- **Higher fanout.** Internal nodes carry only keys + child pointers — no values — so more keys fit per page. For 4 KB pages with 8-byte keys + 8-byte child pointers, a B+ tree internal node holds ~ 256 keys while a same-page B-tree (with embedded values, say 64 bytes each) holds ~ 50 keys.

B-tree vs B+ tree — side by side

Feature	Classic B-tree	B+ tree
Where values live	internal + leaves	leaves only
Routing keys	not duplicated	duplicated at leaf
Internal-node fanout	moderate	high (no values)
Range scan	in-order traversal $O(R \log_K n)$	leaf walk $O(R)$
Sequential scan	re-descend per row	one leaf walk
Point-query depth	may stop at internal	always reaches leaf
Production use	rare today	standard everywhere
Insert / delete algorithm	as in §11.3 / §11.5	*identical* (with leaf-link bookkeeping)

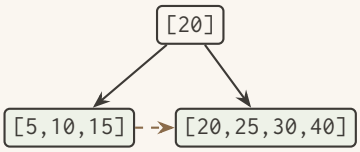
The picture: same conceptual tree shape, but data + links live at the leaf level only.

Classic B-tree



value for "20" lives in the root node

B+ tree



"20" duplicated in the leaf (internal copy is routing only)

Aside

Why B+ wins for databases. A point query on a B+ tree always descends to a leaf (one extra cache line vs the classic B-tree's optional early-stop), but the win comes elsewhere: a range query `SELECT * WHERE x BETWEEN 100 AND 200` descends once to find $x = 100$ and then walks the leaf linked list emitting tuples until $x > 200$. No re-descent per tuple, $O(R)$ in the result-set size R, and the leaf walk is sequential disk access (the OS prefetcher loves it). On a classic B-tree, the same query is $O(R \log_K n)$ because in-order successor traversal climbs back up the tree between leaves.

Example

Worked deletion exercising the 4-case CLRS taxonomy (§11.4). We start with a populated B-tree of order $K = 4$ (min-degree $t = 2$) and delete a sequence of keys to hit each case.

Starting tree:

```
                    [20, 40]
                    /  |  \
[5, 10, 15]  [25, 30, 35]  [45, 50, 55]
```

Delete 50 (Case 1 — key in leaf, leaf has $\geq t$ keys). The leaf `[45, 50, 55]` has 3 keys $\geq t = 2$. Direct delete + shift:

```
                    [20, 40]
                    /  |  \
[5, 10, 15]  [25, 30, 35]  [45, 55]
```

Delete 20 (Case 2a — internal node, predecessor side has spare). Key 20 is at slot 0 of the root. Left child `[5, 10, 15]` has 3 keys ($\geq t$). Find predecessor (`btreeGetMaxKey` of left subtree = 15), swap into root, then recursively delete 15 from the leaf (Case 1):

```
                    [15, 40]
                    /  |  \
[5, 10]  [25, 30, 35]  [45, 55]
```

Delete 5 (Case 3a — key not in current node, descent child needs a key from sibling). Root `[15, 40]` doesn't contain 5; descend left into `[5, 10]`. That child has $t - 1 = 1$ keys? No — it has 2 keys, so the descent is safe. (We're at the floor's threshold. Some implementations preemptively rotate here; CLRS waits until the child is at exactly $t - 1 = 1$ key before fixing.) Continue: in the leaf, delete 5 (Case 1):

```
                    [15, 40]
                    /  |  \
[10]  [25, 30, 35]  [45, 55]
```

Delete 10 (Case 3a — now the leaf is at the floor). Descent reaches root `[15, 40]`; left child is `[10]` with $t - 1 = 1$ key — below the safe threshold for descent. Sibling `[25, 30, 35]` has $\geq t$ keys, so **rotate** the separator 15 down into the left child and pull the smallest sibling key 25 up to the root:

```
                    [25, 40]
                    /  |  \
[10, 15]  [30, 35]  [45, 55]
```

Now descend (safe) into the left child and delete 10 (Case 1):

```
                    [25, 40]
                    /  |  \
[15]  [30, 35]  [45, 55]
```

Delete 25 (Case 2c — internal node, both adjacent children minimal). Key 25 is at slot 0 of the root. Left child `[15]` has $t - 1 = 1$ key, right child `[30, 35]` has 2 keys — not both minimal, so this falls into **Case 2b** (predecessor minimal but successor has spare). Use the successor: minimum of `[30, 35]` = 30, swap into root, recursively delete 30 from right-sibling leaf (Case 1):

```
                    [30, 40]
                    /  |  \
[15]  [35]  [45, 55]
```

Delete 35 (Case 3b — descent child minimal, both siblings minimal). Root `[30, 40]` doesn't contain 35; descend into middle child `[35]`. That child has 1 key, both immediate siblings (`[15]` and `[45, 55]`) — the right sibling has $\geq t$ keys, so this is actually **Case 3a**: rotate from right sibling. Pull separator 40 down, pull smallest sibling key 45 up:

```
                    [30, 45]
                    /  |  \
[15]  [35, 40]  [55]
```

Now descend safely; delete 35 from middle leaf:

```
                    [30, 45]
                    /  |  \
[15]  [40]  [55]
```

Delete 40 (true Case 3b — all three children at the floor). Root `[30, 45]`; key 40 sits between the two separators, so descent goes into middle child `[40]`. Middle has $t - 1 = 1$, left and right both have $t - 1 = 1$ — both siblings minimal. **Fuse**: pick the left sibling + separator 30, fuse with middle into one node `[15, 30, 40]`. The root loses key 30 and child pointer:

```
                    [45]
                    /  \
[15, 30, 40]  [55]
```

Now descend into `[15, 30, 40]` (3 keys, safe) and delete 40:

```
        [45]
        /  \
[15, 30]   [55]
```

Across 7 deletions we exercised all four cases (1, 2a/2b, 3a, 3b) and never recursed back up the tree. Single downward pass per delete, in line with the preemptive-merge invariant.

6 11.6 Memory hierarchy and B-tree variants

The basic B-tree of §11.1–§11.5 is the foundation; production systems layer on variants that target different points in the memory hierarchy.

2-3-4 trees and red-black trees are the same data structure

Aside

The 2-3-4 ↔ red-black equivalence (CLRS Ch. 13 problem 13-4). Every red-black tree corresponds *exactly* to a 2-3-4 tree under the following rule:

- A 2-node in the 2-3-4 tree becomes a black node in the RB tree (with 2 black/red children handled by the recursion).
- A 3-node becomes a black node with one red child (the red child contributes the second key).
- A 4-node becomes a black node with two red children (the red children contribute the second and third keys).

The black-height of the RB tree equals the height of the 2-3-4 tree; red-black insertion / deletion fixups exactly emulate 2-3-4 split / fusion. AVL is the only major balanced-binary-tree that doesn't have a clean 2-3-4 mapping — it sits in a different equivalence class. This explains why **ch. 9** red-black insert/delete had so many cases (red uncle / black uncle / left/right zig-zag): it's **the 2-3-4 algorithm in disguise**, with the cases being the "which colour pattern correspond to which 2-3-4 node?" decoding.

Cache-oblivious B-trees

Aside

Cache-oblivious B-trees (Bender, Demaine, Farach-Colton 2000). The classic B-tree achieves $O(\log_B n)$ block transfers *when you know* B (the block size — you tune the order K to match). What if your code has to run well across L1, L2, L3, DRAM, NVMe, and disk simultaneously, where each level has a different B? You'd need a different B-tree per level.

The **van Emde Boas layout** gives a single tree structure that achieves $O(\log_B n)$ transfers *for every* B *simultaneously*, without the algorithm knowing B. Recursively split a height-h tree into a top half (height $h/2$) and bottom half (height $h/2$), laying them out contiguously in memory; recurse to the leaves. The analysis is non-trivial but the layout is what every modern "cache-oblivious" data structure uses.

Practical impact: production engines (e.g., Google's LevelDB internals, recent academic OLTP work) sometimes prefer cache-oblivious layouts to portable B-trees because they don't have to be retuned per platform.

LSM-trees: the write-optimized alternative

Aside

Log-Structured Merge trees (O'Neil, Cheng, Gawlick, O'Neil 1996). B-trees are read-optimized: a random read costs $O(\log_B n)$ block transfers, but a random insert costs the same because the tree must be modified in place at the leaf. For write-heavy workloads (logging, time-series, key-value stores serving high writes), the in-place update is the bottleneck.

LSM-trees buffer writes in memory in a sorted in-memory structure (often a skip list or red-black tree, called the *memtable*); when the memtable hits a size threshold, it is flushed to disk as a sorted immutable file (an *SSTable*). Reads check the memtable first, then check on-disk SSTables in reverse-chronological order. Background compaction merges SSTables to keep read amplification bounded.

LSM-trees have higher write throughput (sequential writes are roughly $100\times$ faster than random writes on disk and SSDs) and worse read latency (multiple SSTables may need to be checked per key). Production: **LevelDB**, **RocksDB**, **Cassandra**, **HBase**, **ScyllaDB**. The B-tree-vs-LSM choice is the canonical read-vs-write trade-off in storage system design.

Other variants worth pointing at

Aside

Fractal trees / B^ε-trees (Brodal & Fagerberg 2003, Bender et al. 2007): an asymptotic improvement on B-trees that trades some read latency for far better insert throughput, by buffering inserts at internal nodes. The basis of the **TokuDB** / **TokuMX** storage engine (now part of Percona).

Bw-trees (Levandoski, Lomet, Sengupta 2013): a lock-free B+-tree variant designed for many-core in-memory databases. The basis of **Microsoft Hekaton**'s in-memory engine and influential in modern OLTP storage.

Copy-on-write B-trees: every modification creates a new path from leaf to root, leaving the old tree untouched. **Btrfs** and **ZFS** are built on this, and it underpins the snapshot / clone primitives those filesystems expose.

7 11.7 Production references and further reading

Aside

Where B-trees show up in real systems.

- **SQLite.** B+ tree, default page size 4096 bytes (configurable 512–65536). Both table indexes and the rowid B+ tree storing the table itself.
- **PostgreSQL.** `btree` access method (the default for any `CREATE INDEX` that doesn't say otherwise); also has `gist` (R-tree-flavored, for spatial data) and `gin` (inverted index, for full-text and array containment). The `btree` implementation is a B+ tree with concurrent-access mechanics (Lehman–Yao 1981 right-link technique).
- **MySQL InnoDB.** Clustered B+ tree on the primary key (table data lives at the leaves); secondary indexes are separate B+ trees that point at primary-key values. Default page size 16 KB.
- **Linux ext4.** `htree` variant for directory entries (an HTree is a B-tree-like hashed-index structure — enough fanout to keep large directories fast).
- **Btrfs.** Copy-on-write B+ trees throughout the filesystem (extent allocation, file-system tree, subvol-

umes, checksums). The "B" is for B-tree.

- **git packfile (.idx).** *Not* a B-tree — a sorted-array index with a small fanout table at the front for log-time lookup. Worth contrasting: git's access pattern is read-mostly, append-only, and doesn't need range queries; the simpler structure is the right call.
- **LSM-tree-on-top engines. LevelDB**, **RocksDB**, **Cassandra**, **HBase**, **ScyllaDB**. Don't use B-trees on disk (write-optimised), but the in-memory memtable is often a skip list or red-black tree, and on-disk SSTables are sorted-string-table files that get B-tree-like indexes themselves.

Canonical references.

- **CLRS Ch. 18** (B-Trees). The textbook treatment; minimum-degree t convention, B-TREE-SEARCH / SPLIT-CHILD / INSERT-NONFULL, the 4-case deletion taxonomy. The version cited throughout this chapter.
- **Goetz Graefe, "Modern B-Tree Techniques"** (Foundations and Trends in Databases, 2010). 200-page monograph covering production-engineering details: concurrent access, write-ahead logging integration, bulk-loading, prefix B-trees, and partitioning. The industry reference.
- **Knuth, TAOCP vol. 3 §6.2.4.** The original treatment of multiway trees, including Knuth's bottom-up insert (the "lazy alternative" warnbox in §11.3).
- **Bender, Demaine, Farach-Colton, "Cache-Oblivious B-Trees"** (FOCS 2000). The van Emde Boas layout result referenced in §11.6.
- **Bayer & McCreight, "Organization and Maintenance of Large Ordered Indexes"** (1972). The original B-tree paper.

Key idea

Chapter takeaway. A B-tree is a balanced search tree where *balance via fanout, not branching* is the operating principle: each non-root node carries between $\lceil K/2 \rceil - 1$ and $K - 1$ keys, which gives $h = O(\log_K n)$ and ties tree height directly to the memory hierarchy — one block per level by design. The chapter unifies under four sub-algorithms, all single-pass:

- **Search** (§11.2): in-node scan + single descent per level. CPU $O(\log n)$ with binary in-node search; disk $O(\log_K n)$ regardless. Same primitive as classic BST search, applied per node.
- **Preemptive-split insert** (§11.3): split full nodes *on the way down*; the descent invariant "current node has room" is preserved. `btreeSplitChild` + `btreeInsertNonFull` together — one downward pass, no parent pointers, no second pass.
- **Preemptive-merge remove** (§11.4–§11.5): mirror — rotate-or-fuse on the way down (cases 3a / 3b of the CLRS taxonomy) so the leaf to delete from always has $\geq t$ keys. Internal-node deletion (case 2) replaces the key with a predecessor or successor and recurses into a leaf; case 2c fuses the two children when both are minimal.
- **Rotation + fusion** (§11.4): the two constant-cost local restructuring primitives both insert and remove rely on. Rotation is the inverse of itself (left $\leftrightarrow$ right); fusion is the inverse of split. Every B-tree modification is one of these four.

With K in the hundreds, $\log_K n \leq 5$ for billions of keys — B-tree lookups are essentially $O(1)$ on any realistic workload, and that's why every serious storage system in production uses one.

Looking ahead. Chapter 12 covers the *set ADT* in full. The "does this set contain x?" / "insert x" / "remove x" / "iterate in order" interface is what B-trees and B+ trees implement on disk; `std::set` / `std::map` use red-black trees (the in-memory dual — recall §11.6's 2-3-4 $\leftrightarrow$ red-black equivalence),

Java's `TreeMap` likewise, and SQLite / PostgreSQL / InnoDB use B+ trees. The chapter will frame these as implementation choices for the same ADT. **Forward direction beyond cs-300**: the variants in §11.6 (cache-oblivious B-trees, LSM-trees, fractal trees, Bw-trees) are where modern storage-system research lives, and Graefe's monograph (§11.7) is the entry point.